

User Need and AI Contribution

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of capstone requirements and corpus.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Multimodal Evidence. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 19: User, Need, and AI Contribution

AI should solve a defined user problem better than a simpler alternative.



CREATOR MISSION

Compare AI and non-AI designs for a study need.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall System evaluation sets. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

By the end, I can explain user need and ai contribution using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use user, need, and ai contribution to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A grounded-assistant capstone begins with bounded requirements and a versioned permitted corpus. A baseline can be a manual lookup process using the same evidence.

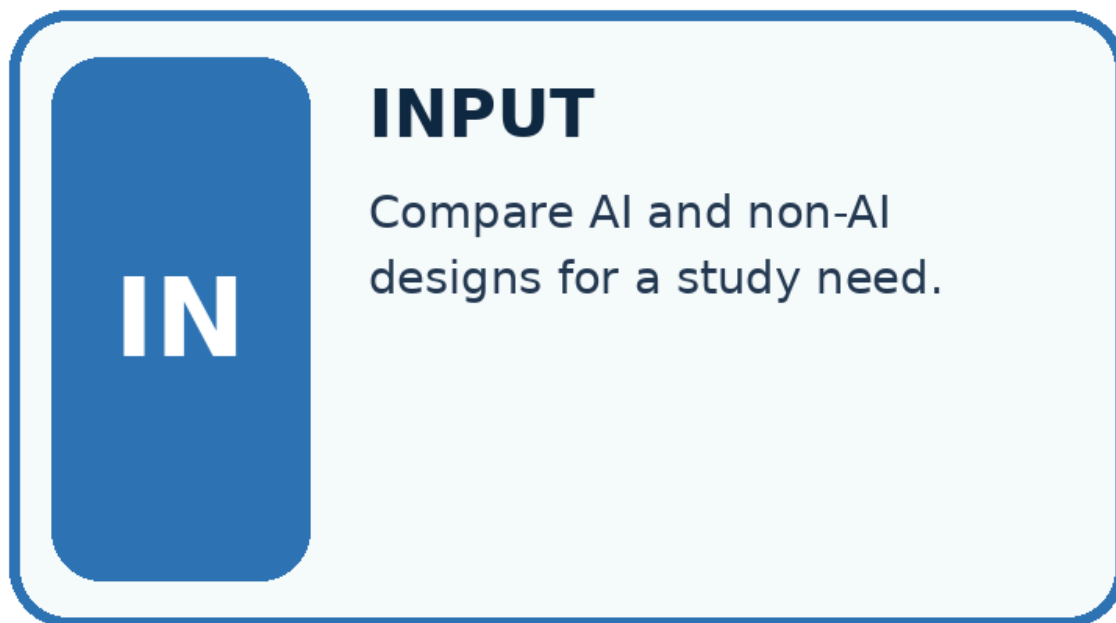
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

AI should solve a defined user problem better than a simpler alternative.





EVIDENCE

Compare AI and non-AI designs for a study need.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Specify the user question, source authority, access, output/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

Capstone requirements and corpus — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.

Check or calculation: Specify the user question, source authority, access, output/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

Conclusion and limit: Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: The pupil asks for photosynthesis, which G3 does not contain.

Prediction: The assistant reports that the definition is unavailable in its approved glossary.

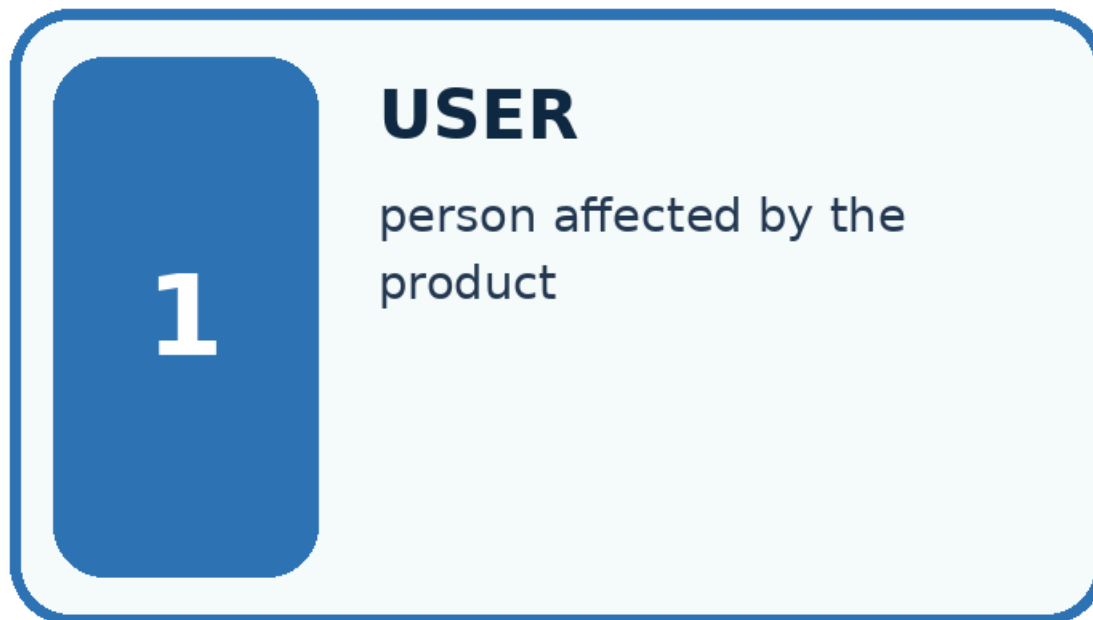
Reason: A helpful interface cannot expand its authorised evidence by guessing.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

NEED

observable problem or goal

3

AI CONTRIBUTION

task where model behavior
adds value

4

SIMPLER ALTERNATIVE

non-AI way to meet the need

Word · My own example · Not this / boundary

Word: user

My own example: The person who interacts with or relies on a system.

Not this / boundary: _____

Word: need

My own example: A problem or desired outcome for the intended user.

Not this / boundary: _____

Word: AI contribution

My own example: The specific part of a task an AI system is intended to help perform.

Not this / boundary: _____

Word: simpler alternative

My own example: A less complex approach that may meet the same user need with fewer resources or risks.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Requirement

Meaning: A declared user-visible condition the system must satisfy

Word: Corpus manifest

Meaning: An inventory of source IDs, versions and access roles

Word: Baseline procedure

Meaning: A credible reference process on the same task

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 19 • WORKED EVIDENCE

- 1 Need: turn approved notes into practice questions.
- 2 Simple template can create fixed question types reliably.
- 3 AI may add varied wording but also unsupported details.
- 4 Use AI only where benefit exceeds added verification and safety cost.

1. Need: turn approved notes into practice questions.
2. Simple template can create fixed question types reliably.
3. AI may add varied wording but also unsupported details.

4. Use AI only where benefit exceeds added verification and safety cost.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.

Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: User Need and AI Contribution

Define a useful reading helper before choosing its AI features.

The facts we will use

A pupil needs the meaning of unfamiliar words in an approved glossary. G3 defines habitat as the place an organism lives; no other sources are allowed.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

User Need and AI Contribution

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the need

The pupil wants a glossary meaning, not a made-up example presented as a fact.



2. Choose contribution

The assistant locates the glossary entry and restates it in clear language with G3.



3. Bound the service

An absent word produces "not in this glossary"; a direct glossary search is the baseline.

CHANGE ONE THING

The pupil asks for photosynthesis, which G3 does not contain.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Name the need

What happens and why: The pupil wants a glossary meaning, not a made-up example presented as a fact.

Step: 2. Choose contribution

What happens and why: The assistant locates the glossary entry and restates it in clear language with G3.

Step: 3. Bound the service

What happens and why: An absent word produces “not in this glossary”; a direct glossary search is the baseline.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A requirements plan identifies intended value; it does not demonstrate actual pupil benefit.

Read the labelled picture: What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: The pupil asks for photosynthesis, which G3 does not contain. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Adding AI automatically makes a product more useful. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help

used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A complete capstone plan is not evidence that the implementation meets it; evaluate after building.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone requirements and corpus”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone requirements and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Interview one learner, write a need statement, sketch AI and non-AI flows, and compare value, risk, cost, and testability.

Interview one learner, write a need statement, sketch AI and non-AI flows, and compare value, risk, cost, and testability.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Justify exactly which Study Buddy stage uses AI and which stages remain deterministic.

THIS WEEK’S CHECKPOINT Justify exactly which Study Buddy stage uses AI and which stages remain deterministic.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

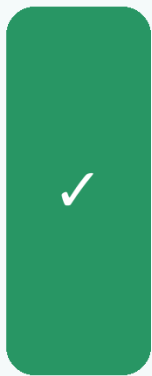
For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Compare AI and non-AI designs for a study need.

CHALLENGE Compare AI and non-AI designs for a study need.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 19 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Learners struggle to locate definitions in an approved chapter; a proposed chatbot answers everything.

The question

Narrow the need and AI contribution.

Worked explanation

Help locate and explain chapter definitions with references and unsupported-question handling. Compare a simple index or search first.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Choose a narrow, testable contribution

Evidence or stage · Value or result

Evidence or stage: Need

Value or result: Find chapter definitions

Evidence or stage: Simple baseline

Value or result: Index or search

Evidence or stage: AI help

Value or result: Explain with chapter references

Evidence or stage: Limit

Value or result: Handle unsupported questions

Learners struggle to locate definitions in an approved chapter; a proposed chatbot answers everything.

Narrow the need and AI contribution.

Help locate and explain chapter definitions with references and unsupported-question handling. Compare a simple index or search first.

14. Try a new case independently

Independent case: Apply a new case independently

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice

B. Define eligible corpus, baseline and a missing-evidence test.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 19 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A student needs the exact due date from a course notice. Should the design start by generating a plausible date or locating the notice? Explain the AI contribution.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Justify exactly which Study Buddy stage uses AI and which stages remain deterministic.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Students cannot find definitions in a 20-page handout. Compare an index with a grounded assistant. Define a success measure and a failure case for both.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Define a useful reading helper before choosing its AI features.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of capstone requirements and corpus

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of capstone requirements and corpus

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or

assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Define a useful reading helper before choosing its AI features.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? An absent word produces “not in this glossary”; a direct glossary search is the baseline.

1. Bound the service
2. Name the need
3. Choose contribution

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Bound the service An absent word produces “not in this glossary”; a direct glossary search is the baseline. Read the picture in order. An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: The pupil asks for photosynthesis, which G3 does not contain. Which conclusion is supported by the supplied facts?

1. The assistant reports that the definition is unavailable in its approved glossary.
2. This one example guarantees correct results for every future input.

3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The assistant reports that the definition is unavailable in its approved glossary. A helpful interface cannot expand its authorised evidence by guessing.

Repair a missing building block

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the mechanism and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Verify Numbers and Dates (ai-3/16). Goal: Explain the mechanism and terminology of claim support and citation evaluation. Short practice: Which part can be corrected immediately from existing evidence, and which needs follow-up?. Return to this week and explain what you changed.

Review Multimodal Evidence (ai-3/13). Goal: Explain the mechanism and terminology of speech pipelines and multimodal evidence. Short practice: Why were both “exactly three” and “exactly five” unsupported before opening the box?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.